

Note Tracking Application

Secure AI-Powered Note Management System



Overview

The Secure AI-Powered Note Management System is a modern web application designed to help users create, organize, secure, and share digital notes. The system combines AI-powered automation, secure collaboration, and user-controlled version history to provide an efficient and professional note management experience suitable for assignments, long-term projects, and personal productivity.



Unique Features



User-Defined Manual Versioning (Git-Style)

Users can manually create version snapshots with custom labels such as **Draft v1**, **Final Review**, or **Submission Version**. This allows clear milestone tracking and avoids unnecessary automatic versions.



AI Smart Assistance

AI helps users by:

- Suggesting relevant tags
- Predicting categories
- Generating summaries for long notes

This reduces manual organization effort.

Dual Security Layer

The system provides enhanced security through:

- Encrypted note storage
 - Optional folder-level password protection
 - Private or shared visibility control
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Role-Based Collaboration

Shared notes support permission-based access:

- **Owner** – Full control
- **Editor** – Can modify notes
- **Viewer** – Read-only access

Activity tracking is included for collaboration transparency.

Version Timeline Interface

A timeline-based version history allows users to:

- View previous versions
 - Preview old content safely
 - Restore versions with one click
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All System Features

User Management

- User registration and login
 - Secure authentication
 - Profile management
 - Friends system for quick sharing
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Note Management

- Create, edit, delete, and view notes
 - Markdown editor support
 - Autosave draft system
 - Trash system with restore option
 - Archive notes
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Note Organization

- Nested folders (sub-folders)
 - Tags and categories
 - Multi-tag filtering
 - Favorites and pinned notes
 - Search functionality
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AI Features

- Smart tag suggestion

- Smart category prediction
 - AI summarization
 - Manual AI trigger option
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Security & Privacy

- Encrypted note storage
 - Folder-level password protection
 - Private or shared visibility settings
 - Access control validation
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Collaboration Features

- Share notes with selected users
 - Permission-based access control
 - Activity logs showing edits and actions
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Version Control Features

- Manual “Save Version” action
- Custom version labels and comments
- Version history sidebar
- View previous versions without overwriting current work
- Restore any previous version as the latest snapshot

Activity & Notification Features

- Track edits and sharing events
- Version restore history
- Collaboration activity updates

General System Capabilities

- Responsive interface
- Efficient search and filtering
- Stable and scalable structure
- Smooth editing experience